Scalability & Future Plan

Date	3 July 2026
Team ID	XXXXXX
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a limited historical dataset	May reduce prediction accuracy for unseen conditions	High
2	Supports prediction for one location at a time	Cannot monitor multiple regions simultaneously	Medium
3	Local deployment only	Limited accessibility for end users	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users	Deploy on IBM Cloud/AWS with load balancing to support many concurrent users
2	Data Storage	Uses a local CSV dataset	Store data in a cloud database such as MySQL or PostgreSQL
3	Performance	Prediction runs on a local machine	Optimize the model and deploy using scalable cloud services
4	Security	Basic input validation	Implement HTTPS, user authentication, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time weather APIs for live flood prediction	2–3 months	More accurate and up-to-date predictions
Phase 3	Add GIS map visualization and location-based flood alerts	4–6 months	Better monitoring and easier decision-making
Phase 4	Develop Android/iOS mobile application with notification alerts	6–12 months	Improved accessibility and faster emergency response